Duty at home

DM12 · Projection and intersection · 55 min.

In an orthonormal space coordinate system, the point is considered $M(3;0;0)$ and plans $\mathcal{P} : x+2y+2z=9$ and $\mathcal{Q} : x-y=0$.

1. Give the coordinates of a normal non-zero vector to $\mathcal{P}$.
2. Determine coordinates of the orthogonal project H of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$.
3. Show that $\mathcal{P}\cap\mathcal{Q}$ is a straight and give a parametric representation.
4. Item H Does it belong to this right?
5. Determine coordinates of the point K of this right whose rating is zero. MKthen compare MK and MH.